

A Unified Framework for Designing Swarm Robots Using a Model-Based Systems Engineering Approach

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ABSTRACT Multi-robot applications are difficult to validate when system models, simulation assets, and robot software evolve separately. This paper presents a model-based systems engineering (MBSE) workflow that connects SysML artifacts to an executable ROS 2 implementation for a two-TurtleBot4 target-acquisition mission. The workflow defines requirements, behavioral views, software architecture, hardware architecture, and data-flow views, then maps them to ROS 2 nodes, topics, launch files, DDS network settings, and validation metrics. The implemented case study uses exactly two canonical robot identities, `/robot1` and `/robot2`, while the physical TurtleBot4 Lite deployment boundary records the required FastDDS, `ROS_DOMAIN_ID`, Simple Discovery, and `/cmd_vel_unstamped` interfaces. A coordinator assigns `/robot1` as leader, assigns `/robot2` a dynamic follower pose behind the leader, publishes observable task state and events, and records the result in CSV metrics. Fifteen stable Gazebo simulation trials over five target positions on the saved lab occupancy map completed successfully, with a mean completion time of 7.603 s, no role switches, and a minimum observed inter-robot separation of 0.321 m. A physical deployment check confirmed the canonical odometry and command interfaces while identifying lidar scan streaming as a remaining deployment issue. The contribution is not a general swarm benchmark, but a traceable sim-to-real workflow whose scope is deliberately aligned with the available two-robot platform.

INDEX TERMS Model-based systems engineering, SysML, TurtleBot4, ROS 2, swarm robots, sim-to-real validation

I. INTRODUCTION

Model-based systems engineering (MBSE) is increasingly used to manage the interaction between requirements, architecture, behavior, and verification in complex engineered systems [1], [2]. Robotics brings the same integration problem into a smaller but more dynamic setting: the model must remain consistent with sensors, computation, communication, motion control, simulation, and hardware deployment. This consistency is especially important for multi-robot systems because a coordination error may be caused by an ambiguous requirement, an incomplete architecture model, a ROS 2 namespace or `ROS_DOMAIN_ID` mismatch, a simulator limitation, or a hardware safety constraint.

This paper studies that problem through a deliberately bounded case study: two TurtleBot4 robots performing a target-acquisition mission. One robot is assigned as the attacker and moves toward the target. The second robot is

assigned as the supporter and moves to an offset pose near the same target. The number of robots is fixed by the available physical platform, and the paper does not claim validation beyond this two-robot setting.

The central research question is how to keep SysML models useful after the transition from conceptual design to executable ROS 2 code. Rather than treating SysML diagrams as documentation only, the proposed workflow uses them as traceability anchors for the implementation: requirements are mapped to nodes, topics, launch files, runtime events, DDS settings, and metrics. The same canonical robot identities are used in simulation and in the physical deployment adapter so that the design boundary is explicit.

The contributions of this paper are:

- a SysML-based workflow that connects requirements, activity, structure, software architecture, hardware architecture, and data-flow views for a two-TurtleBot4

mission;

- a ROS 2 implementation that realizes the modeled behavior through `/robot1` and `/robot2` canonical identities, observable roles, target goals, task state, and metrics;
- a sim-to-real deployment boundary in which simulation is executable by default and physical motion commands require explicit enablement;
- repeated simulation trials and a physical deployment check showing that the modeled requirements can be traced to executable ROS 2 evidence.

The rest of the paper is organized as follows. Section II reviews related work on swarm robotics, MBSE, SysML, and ROS-based systems. Section III presents the proposed MBSE-to-ROS 2 workflow. Section IV describes the two-TurtleBot4 case study and validation results. Section V concludes the paper and identifies remaining limitations.

II. RELATED WORK

A. SWARM ROBOTICS AND MULTI-ROBOT COORDINATION

Swarm robotics studies how multiple robots can coordinate through task allocation, navigation, communication, and local decision-making. Surveys of swarm robotics tasks emphasize behaviors such as aggregation, flocking, foraging, deployment, collaborative manipulation, and task allocation [3]. From a swarm engineering perspective, these behaviors must be designed, verified, and maintained through systematic processes rather than through isolated controller implementations [4]. Mission-oriented architectures for unmanned swarms further show that reusable mission structure is useful when the same coordination logic must be adapted to different platforms or scenarios [5].

The present work does not attempt to cover all swarm behaviors. It selects one representative coordination problem—role allocation for target acquisition—and uses it to demonstrate traceability from system requirements to executable ROS 2 behavior. This narrower scope makes the validation boundary explicit and keeps the experiment aligned with the available two-TurtleBot4 platform.

B. MBSE AND SYSML FOR EXECUTABLE ENGINEERING

MBSE replaces scattered document artifacts with linked models that capture requirements, structure, behavior, and verification intent [1], [2]. SysML has been used to structure engineering knowledge at multiple abstraction levels, allowing high-level decisions to remain connected to lower-level design elements [6]. Systems-thinking work also emphasizes the transition from problem capture to formal SysML models, reducing information loss between early design and engineering artifacts [7].

Several domains have demonstrated the value of MBSE for architecture and traceability, including aircraft systems [8], [9], satellite communication systems [10], and digital twins for industrial robots [11]. These works show that MBSE is most useful when the model supports downstream analysis or

implementation. The contribution here is to apply the same principle to a small mobile-robot team and to keep the SysML artifacts connected to concrete ROS 2 entities and measured simulation results.

C. ROS-BASED ROBOTIC SYSTEMS

ROS and ROS 2 provide a practical middleware layer for robotic software, but the flexibility of nodes, topics, parameters, and launch files can make architecture-level consistency difficult to maintain. A SysML-based metamodel for ROS systems has been proposed to represent both runtime systems and developer workspaces [12]. Supervisory controller synthesis for ROS-based applications also illustrates the need to connect requirements with executable robotic behavior [13]. This paper follows the same motivation from a traceability perspective: every requirement in the case study is linked to a ROS 2 artifact and a runtime metric.

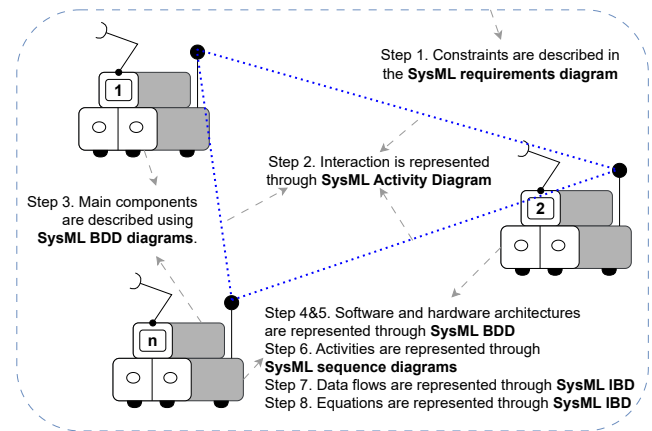


FIGURE 1. Illustrating the use of SysML in the design of swarm robotics

D. SYSML FRAMEWORK FOR SWARM ROBOTS

The objective of using SysML is to provide a standardized method for modeling and integrating the hardware, software, and behavioral aspects of a coordinated robot team (see Figure 1). In this paper, the validated team contains two TurtleBot4 robots. The same modeling pattern can be reused for larger teams, but larger teams are outside the validation boundary of the present work.

- **Modeling language:** SysML is used because it represents requirements, behavior, structure, and inter-component relationships within one consistent notation.
- **Tool independence:** The workflow does not depend on a specific SysML tool; it requires only support for requirement diagrams, block definition diagrams, activity diagrams, sequence diagrams, and internal block diagrams.
- **Modeling conventions:** Requirement identifiers, robot identities, ROS 2 topic names, DDS domain settings, and validation metrics are kept consistent so that each diagram can be traced to an executable artifact.

III. PROPOSED MBSE-TO-ROS 2 WORKFLOW

The proposed workflow links a SysML model to executable ROS 2 artifacts. It is organized around four questions: what must the two-robot system do, how is the mission behavior represented, which software and hardware elements realize the behavior, and how is the result verified. Figure 2 captures the top-level requirements, Figures 3–7 provide behavioral and structural views, and Figures 8–10 connect the model to implementation and platform concerns.

A. REQUIREMENT MODEL AND TRACEABILITY

The requirement model defines the validation boundary before implementation. The mission contains exactly two TurtleBot4 robots. They cooperate on one target-acquisition task, use /robot1 as the leader and /robot2 as the follower once fresh odometry is available, publish observable state, validate the behavior in simulation before physical deployment, and preserve a documented mapping to the physical TurtleBot4 Lite network configuration. This early boundary is important because it prevents the model from implying a robot team size or capability that is not available in the experimental platform.

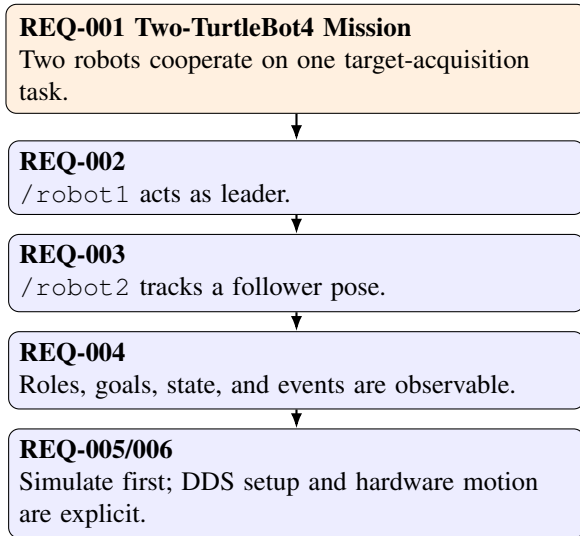


FIGURE 2. High-level requirements for the implemented two-TurtleBot4 case study

Each requirement is later mapped to a ROS 2 artifact: the coordinator node realizes role allocation, goal topics realize target and support assignments, launch files define simulation and physical modes, and CSV metrics provide evidence for verification. This gives the paper a traceable path from requirement to runtime result rather than a diagram-only model.

B. BEHAVIORAL AND STRUCTURAL SYSML VIEWS

The behavioral model uses a general swarm-task taxonomy as context, but selects task allocation, navigation, communication, and metric reporting as the executable subset for this study. Figure 3 shows the broader task model, while Figure 4 shows the activity pattern used by the mission: sense state,

exchange information, allocate a role, execute the assigned action, and report completion.

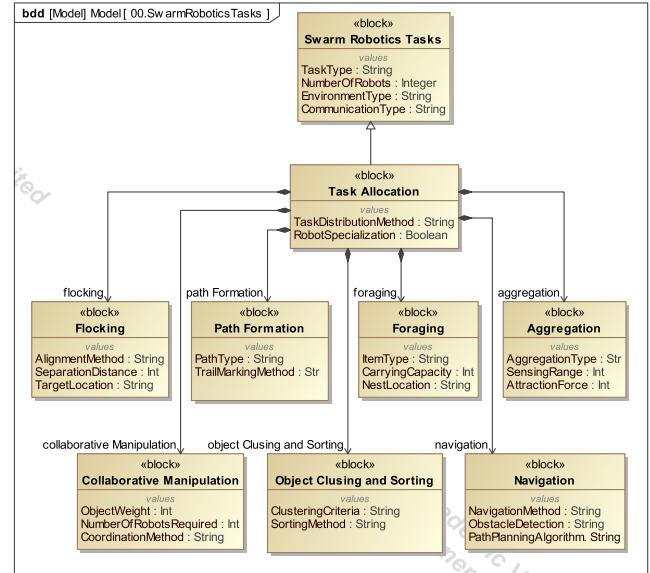


FIGURE 3. Swarm robotics task model used to locate the implemented behavior

The structural view decomposes an individual robot into sensing, communication, control, and actuation responsibilities. In the TurtleBot4 implementation, these responsibilities map to odometry and sensor streams, ROS 2 communication, coordinator decisions, and velocity-command interfaces. Figure 5 gives the block-level view, and Figures 6 and 7 show the interaction sequence used when a robot receives a task, senses its environment, executes a motion command, and reports status.

C. SOFTWARE ARCHITECTURE MAPPING

The software architecture is deliberately small enough to be executable and inspectable. The coordinator consumes target and odometry information, assigns roles, publishes goals, and emits task state. The simulation-only goal follower converts goals into velocity commands. The metrics logger subscribes to task state, events, and odometry to record validation data. In physical deployment, the coordinator and metrics nodes can run without commanding robot motion; direct velocity following is disabled unless explicitly enabled for controlled testing and mapped to the TurtleBot4 Lite /cmd_vel_unstamped interface.

The architecture in Figure 8 is mapped directly to the ROS 2 package. The **ControlSystem** is the paper_swarm_coordinator node. The **MissionState** is represented by /target_pose, /robot*/odom, /swarm/task_state, and /swarm/events. The **MotionControlModule** is represented by the goal follower in simulation and by the external navigation or safety stack when the physical robots are used. For TurtleBot4 Lite hardware, the command endpoint is treated as

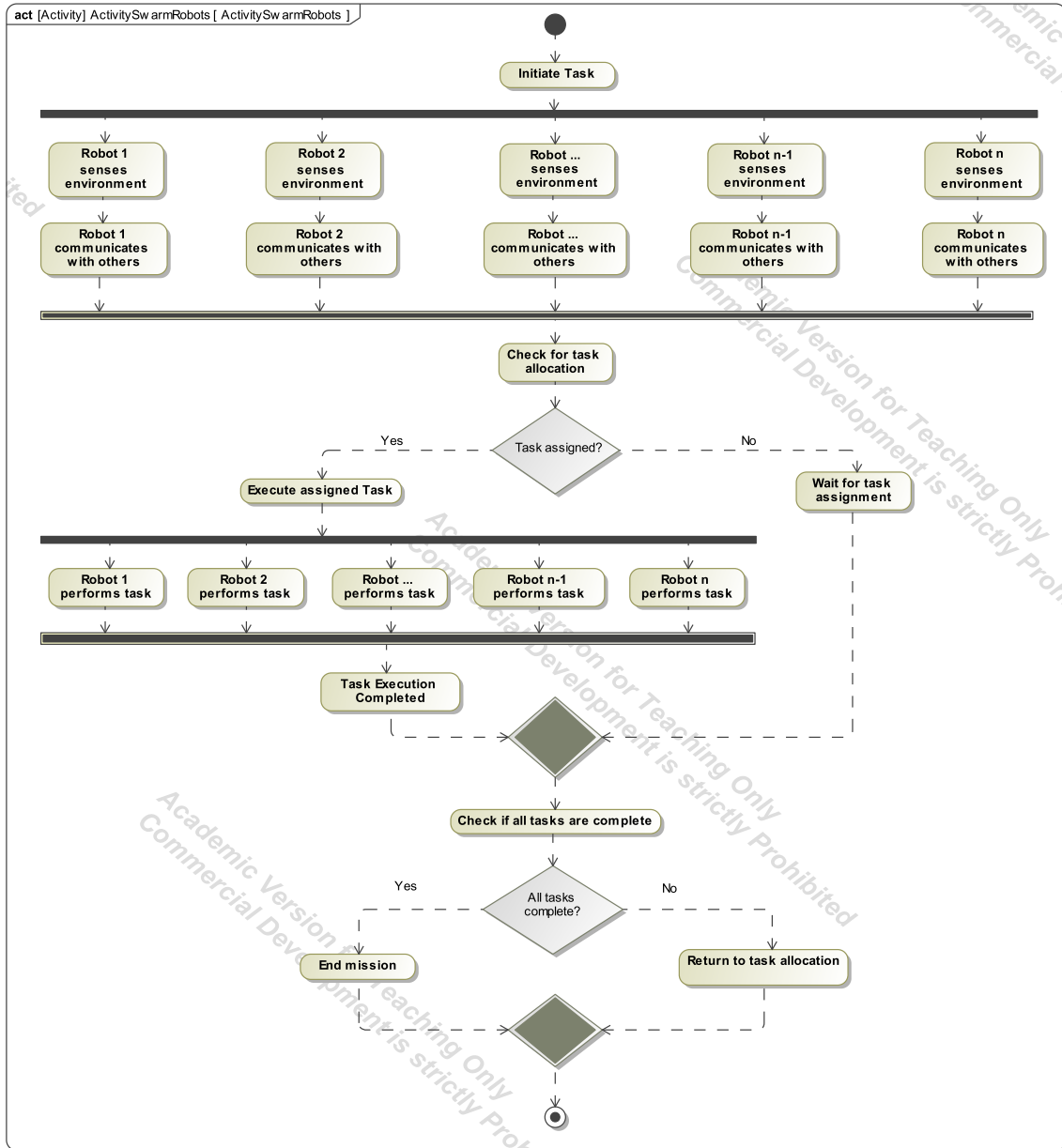


FIGURE 4. Activity model for task allocation and coordinated execution

/robot*/cmd_vel_unstamped after namespace or bridge adaptation.

D. HARDWARE AND DATA-FLOW VIEWS

The hardware model identifies the platform assumptions that the software must respect: differential-drive motion, onboard computing, battery power, lidar/vision/IMU sensing, and wireless communication. The implementation uses TurtleBot4-class interfaces instead of a custom robot design, but the hardware view remains useful because it documents which physical resources are expected by the software architecture.

The Internal Block Diagram (IBD) in Figure 10 describes

how data moves between sensing, processing, communication, power, and actuation elements. The corresponding software data-flow view in Figure 11 connects those platform-level flows to the ROS 2 topics used by the mission. Together, the diagrams justify why odometry freshness, topic identities, DDS domain configuration, and command gating are treated as explicit requirements.

IV. CASE STUDY: TWO TURTLEBOT4 TARGET ACQUISITION

A. EXPERIMENTAL OBJECTIVE

The case study evaluates whether the SysML requirements can be carried through to an executable ROS 2

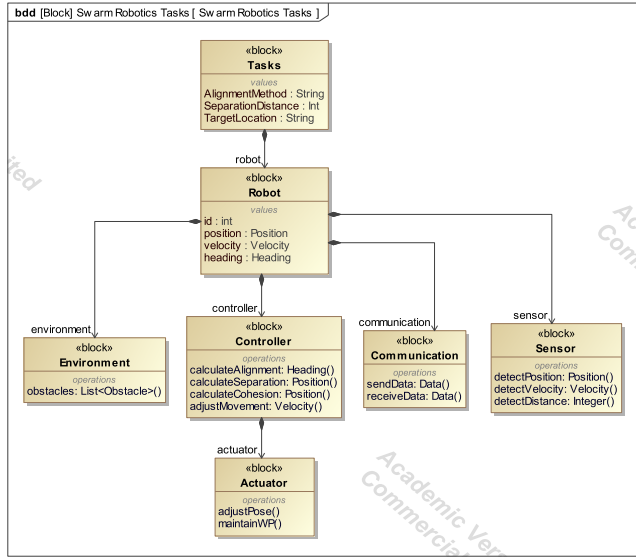


FIGURE 5. Robot-level structural decomposition for sensing, control, communication, and actuation

mission without changing the system boundary. The experimental platform contains two TurtleBot4 robots, represented inside the model and simulator by the canonical identities `/robot1` and `/robot2`. A physical TurtleBot4 Lite is commonly configured as a single robot with Fast-DDS, `ROS_DOMAIN_ID=0`, Simple Discovery, an empty user namespace, and root topics such as `/odom` and `/cmd_vel_unstamped`. Therefore, a two-robot physical deployment must either assign unique namespaces in the shared ROS graph or bridge/remap two single-robot graphs into the canonical `/robot1` and `/robot2` interfaces. The mission is target acquisition: one robot moves to the target as attacker, while the other maintains a nearby support pose. This task is small, but it exercises the essential MBSE links used throughout the paper: requirement decomposition, role allocation, communication, software-to-hardware mapping, simulation validation, and physical deployment gating.

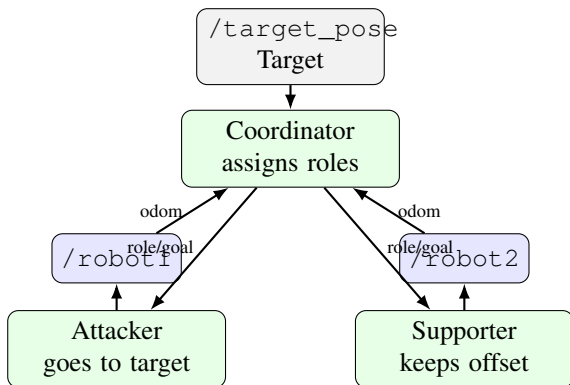


FIGURE 12. Implemented two-TurtleBot4 leader/follower target-acquisition scenario

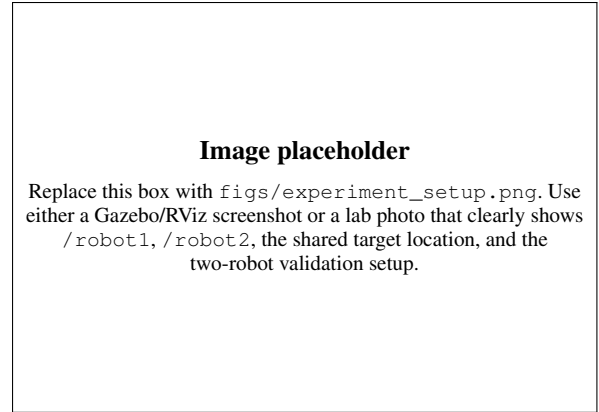


FIGURE 13. Experimental setup for the two-TurtleBot4 target-acquisition validation case.

B. ROLE ASSIGNMENT LOGIC

Let p_T be the target position, p_1 be the latest odometry position for `/robot1`, and p_2 be the latest odometry position for `/robot2`. Each robot is eligible only if its odometry age Δt_i is below the freshness threshold $\tau_{\max}$. The implemented validation case uses a fixed leader-follower assignment:

$$a = \text{/robot1}, \quad s = \text{/robot2}, \quad \Delta t_1, \Delta t_2 \leq \tau_{\max}.$$

The leader goal is the target itself. The follower goal is computed behind the leader along the leader-to-target direction:

$$g_a = p_T, \quad g_s = p_1 - d_f \frac{p_T - p_1}{\|p_T - p_1\|_2},$$

where d_f is the desired follower distance. This logic directly implements REQ-002 and REQ-003 while keeping the experiment deterministic for repeatable validation.

C. RUNTIME SEQUENCE

The runtime sequence follows the activity model in Section III and is limited to target acquisition, role allocation, support positioning, and metric logging:

- 1) A target is published on `/target_pose`.
- 2) Both TurtleBot4 robots publish fresh odometry on the canonical interfaces `/robot1/odom` and `/robot2/odom`, either directly through namespaces or through a physical remapping layer.
- 3) The coordinator verifies that both robot odometry streams are fresh.
- 4) `/robot1` is assigned the leader/attacker role and receives the target as its goal.
- 5) `/robot2` is assigned the follower/supporter role and receives a dynamic pose behind `/robot1`.
- 6) Runtime state and events are published for debugging, traceability, and metric logging.
- 7) Metrics are written to CSV files for validation and regression checks.

D. PAPER REQUIREMENT SET

The SysML requirement model defines the scope of the two-robot case study. Table 1 summarizes the paper-level

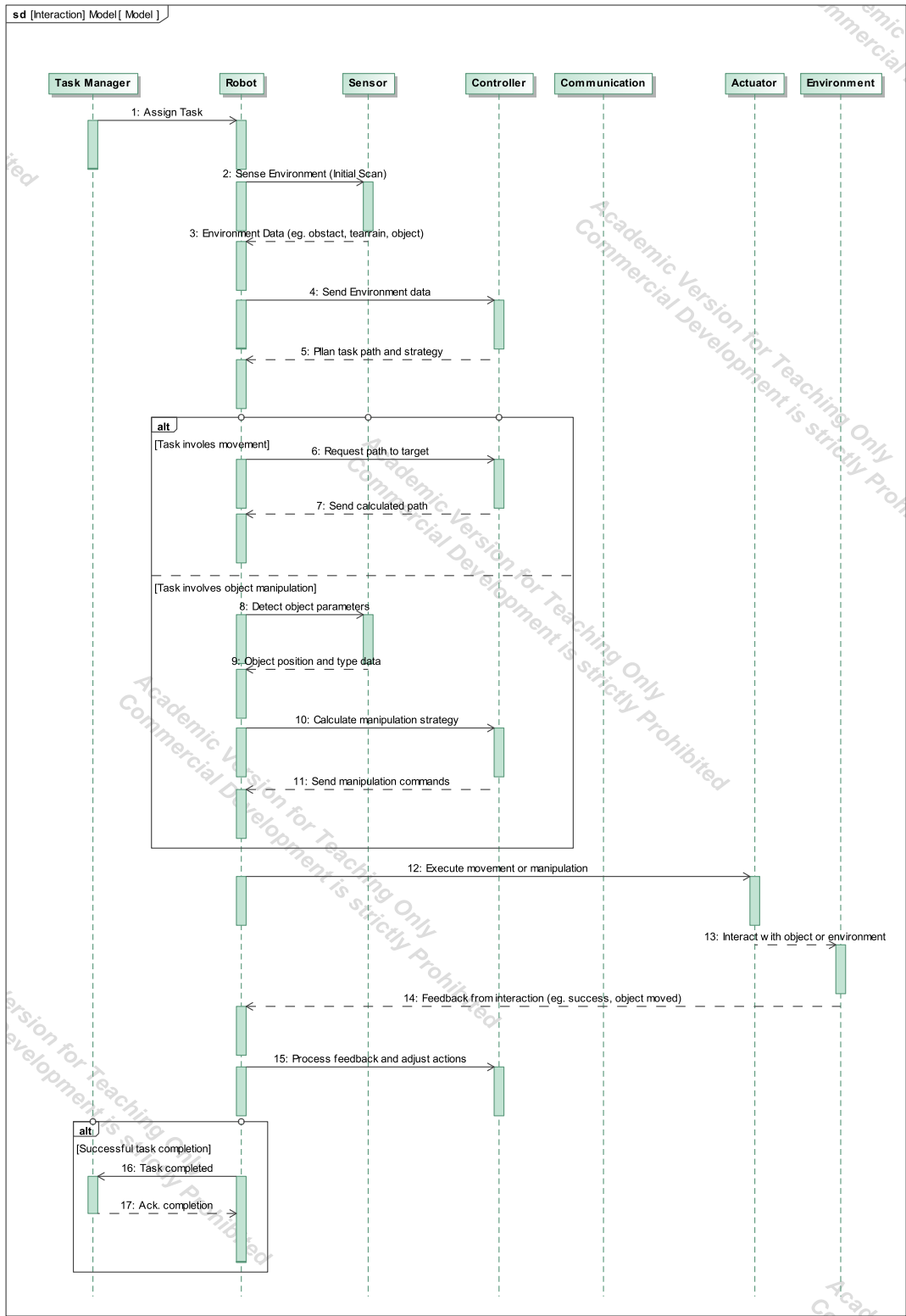


FIGURE 6. Robot interaction sequence for task assignment and sensing

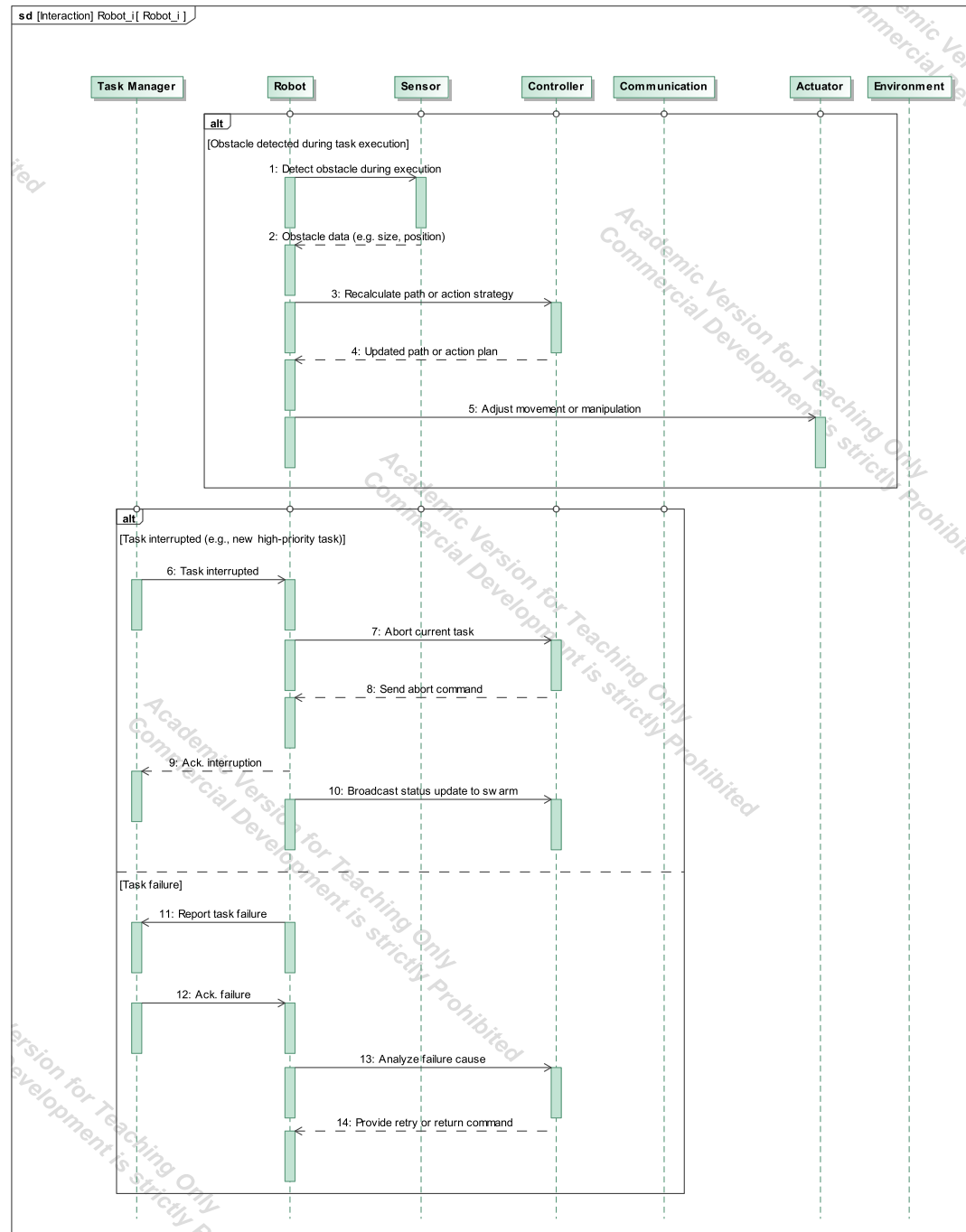


FIGURE 7. Robot interaction sequence for execution, feedback, and status reporting

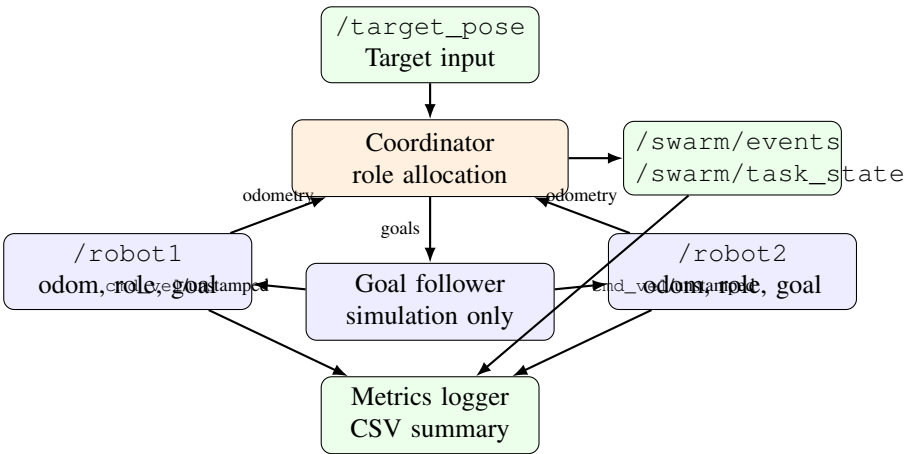


FIGURE 8. Software architecture for the implemented two-TurtleBot4 mission

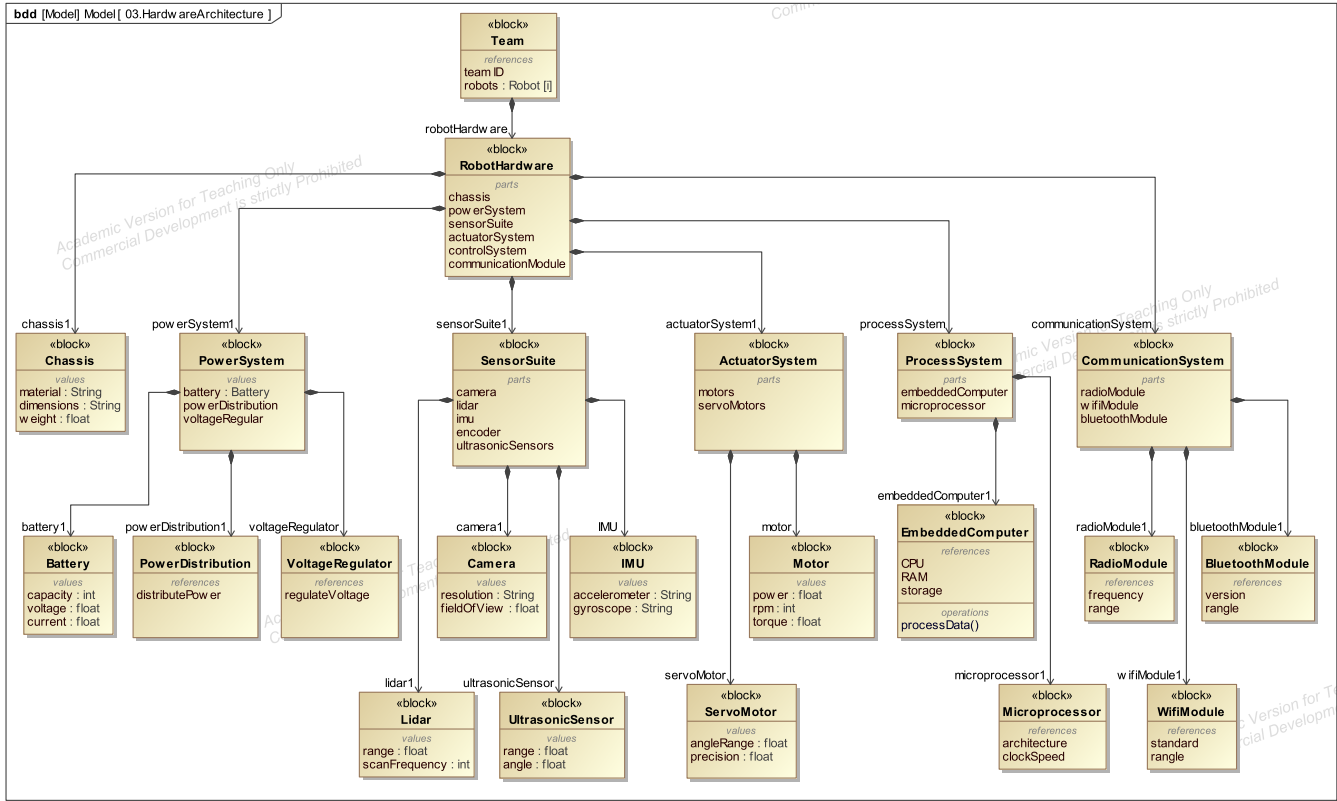


FIGURE 9. Hardware architecture model for one TurtleBot4-class robot

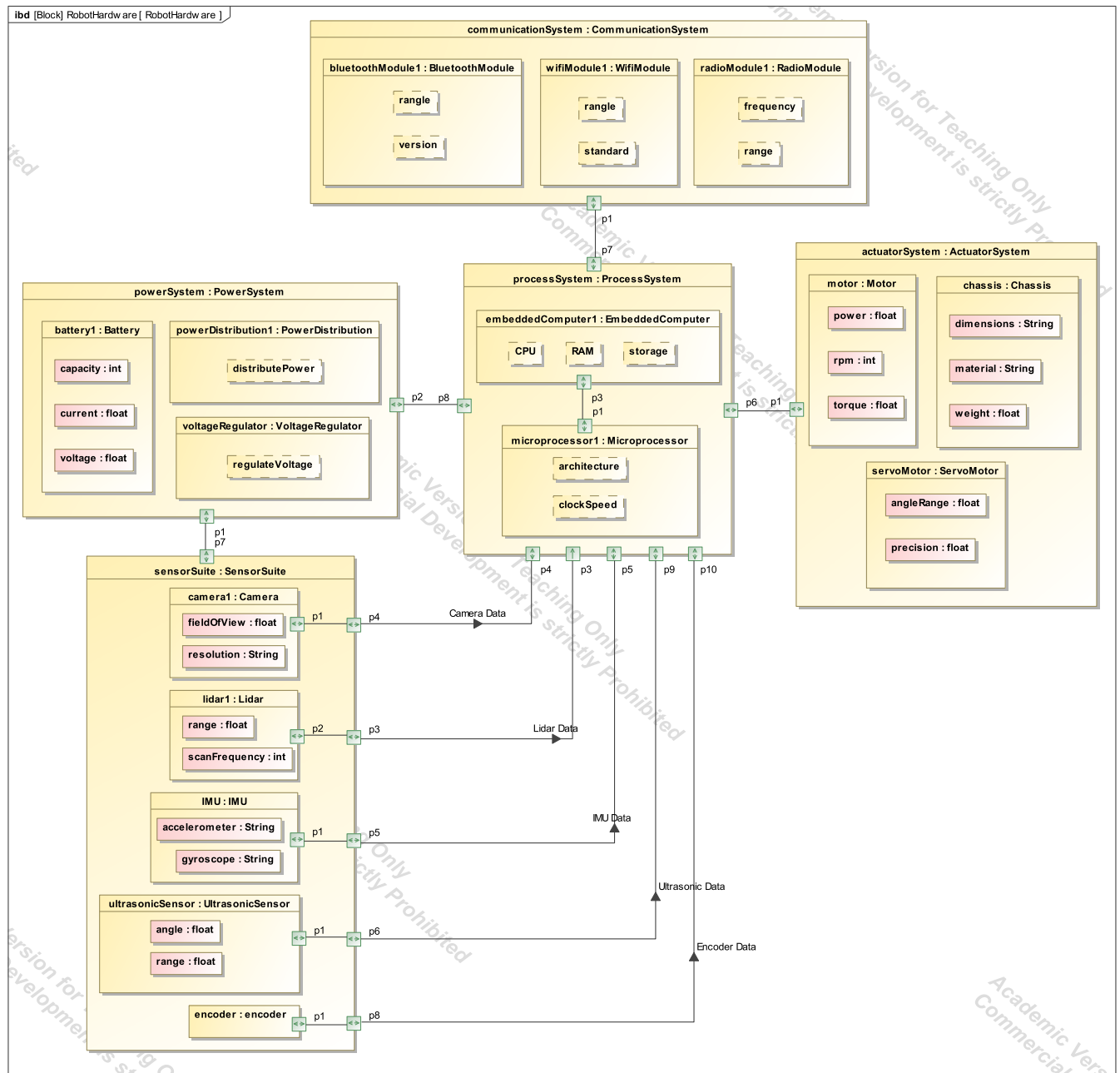


FIGURE 10. Hardware data-flow model using an Internal Block Diagram

requirements used to constrain the simulation and physical validation.

TABLE 1. Paper requirement descriptions

Req.	Description
REQ-001	Two TurtleBot4 robots shall cooperate on one shared target-acquisition task.
REQ-002	/robot1 shall act as the leader/attacker robot during the validation case.
REQ-003	/robot2 shall maintain a follower role by tracking a pose derived from the leader and target geometry.
REQ-004	Robot roles, goals, states, events, and validation metrics shall be observable for experiment analysis.
REQ-005	The two-robot mission shall be evaluated in simulation before physical deployment.
REQ-006	Physical execution shall require explicit DDS configuration and explicit motion enablement before velocity commands are sent to the robots.

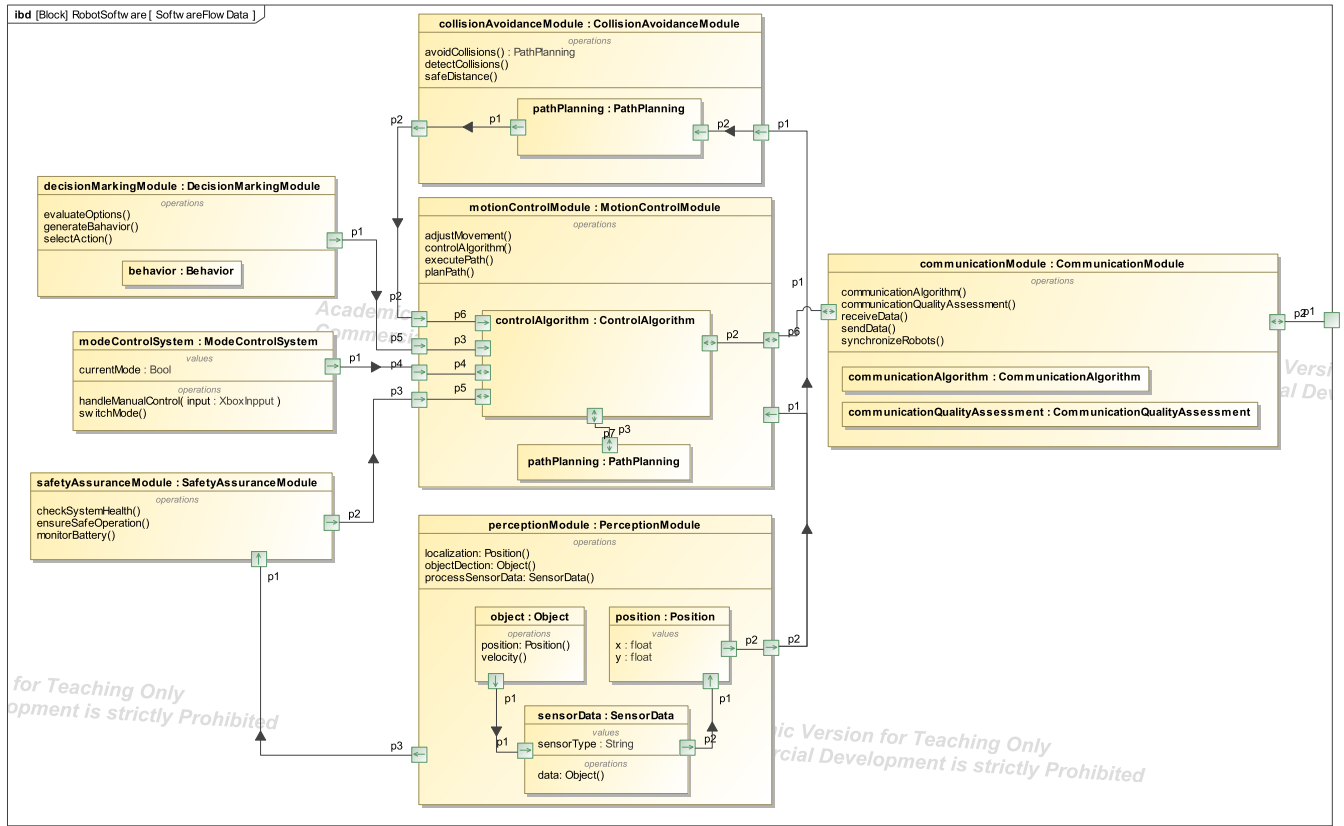


FIGURE 11. Software data-flow view for one robot

E. SIMULATION AND PHYSICAL DEPLOYMENT MODES

Two simulation modes are provided. The stable mode uses lightweight differential-drive physics with TurtleBot4 visual meshes and a Gazebo world generated from the saved lab occupancy map, `maps/lab.yaml`. This mode is the preferred validation path on the current machine because full Gazebo rendering with the official TurtleBot4 stack previously crashed in the renderer. The official mode launches the installed TurtleBot4 simulator twice and is kept server-only by default.

The physical mode does not launch Gazebo and does not command TurtleBot4 motion by default. It starts the coordinator and metrics nodes against the canonical `/robot1` and `/robot2` interfaces. The lab baseline for each TurtleBot4 Lite is FastDDS with `ROS_DOMAIN_ID=0`, Simple Discovery, and no user namespace; in that single-robot form, the robot exposes root topics such as `/odom`, `/scan`, `/oakd/rgb/preview/image_raw`, and `/cmd_vel_unstamped`. Two physical robots cannot both remain unnamespaced in the same ROS graph without topic collisions, so the deployment must either configure unique namespaces or provide an explicit bridge/remap layer before running the two-robot coordinator. Direct velocity following can be enabled only by setting an explicit launch argument during controlled lab testing, and the physical command

topic is `/robot*/cmd_vel_unstamped`. An optional Nav2-based real-robot demonstration can be launched on a saved map, but target selection is limited to safe reachable cells in the map; this optional mode is not used as the quantitative result in this paper. This separation is intentional: the paper validates the model-to-code chain in simulation while preserving a conservative path for physical experiments.

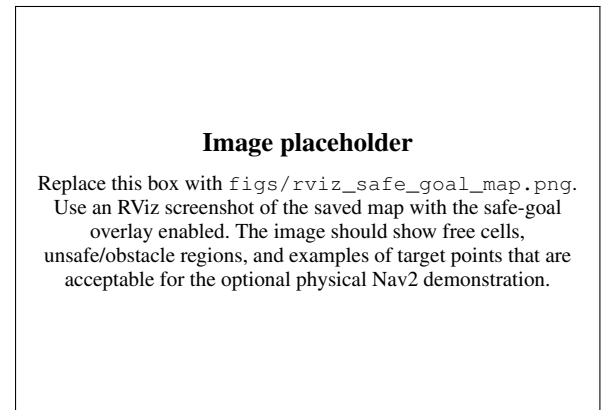


FIGURE 14. RViz map view used to distinguish safe reachable targets from unsafe goal locations in the optional physical Nav2 demonstration.

F. SIMULATION RESULT

A stable simulation campaign was executed with the two TurtleBot4 namespaces and the leader/follower target-acquisition mission on the saved lab map. Five safe target locations were repeated three times, for a total of 15 trials. The targets were (1.422, 0.717), (1.722, 0.717), (1.572, 0.917), (1.522, 1.117), and (1.722, 1.317) m in the map frame. The reported metrics follow common evaluation practice in multi-robot and mobile-robot navigation studies: success rate and path length for task completion, time and travel distance for efficiency, inter-robot separation for safety, and role/formation evidence for coordination [14]–[16]. The target-case identifiers `sim_t01`–`sim_t05` denote the five target coordinates listed in the Target column; the physical table uses the corresponding `real_t01`–`real_t05` identifiers for later real-robot runs on the same map. In the path-length column, R1 denotes the distance traveled by `/robot1`, R2 denotes the distance traveled by `/robot2`, and team denotes the sum of both robot path lengths. The detailed result is shown in Table 2. The result validates the executable chain for target publication, odometry ingestion, role assignment, goal publication, robot motion, task completion, and metric logging.

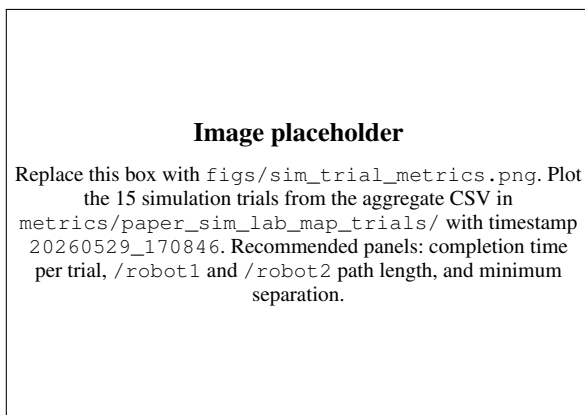


FIGURE 15. Simulation metrics over 15 repeated target-acquisition trials.

Some close-range targets produce zero measured travel for `/robot2` because the follower pose is already within the accepted formation region when the leader reaches the target. This is expected for the compact lab map and is reflected directly in the CSV metrics rather than post-processed away.

G. PHYSICAL DEPLOYMENT CHECK

The physical TurtleBot4 run was treated as a deployment-readiness check before collecting navigation-performance data. The logged readiness checks used FastDDS with `ROS_DOMAIN_ID=0` and confirmed that the odometry streams and command endpoints were discoverable and publishing for both robots, while the lidar scan topics were visible but did not publish `LaserScan` messages within the validation window. Because this readiness state is not yet a physical navigation result, Table 3 is intentionally formatted

as the real-robot result template that will be filled after the same metric set is collected on the physical map.

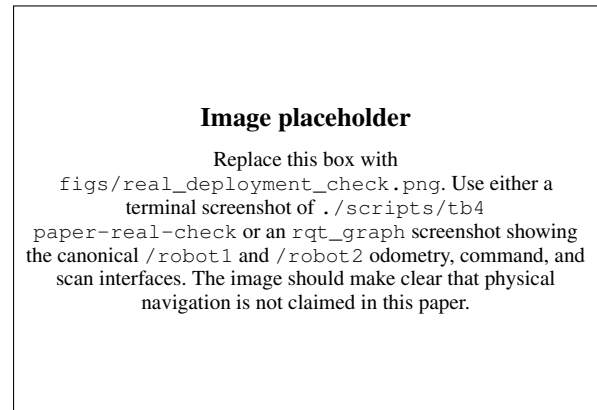


FIGURE 16. Physical deployment check for the canonical two-TurtleBot4 ROS 2 interfaces.

This deployment check is useful because it identifies the remaining physical integration boundary without overstating the experimental result. Until the physical navigation trials in Table 3 are completed, the quantitative evidence in this paper remains the repeated simulation campaign, while the real-robot section records the metric structure that will be used for the final hardware runs.

H. DISCUSSION

The result supports the paper's central claim: the same mission boundary defined in SysML is executable in ROS 2 and produces measurable validation evidence. The 100% success rate over 15 trials shows that the selected two-robot mission can be reproduced across multiple target positions in the stable simulation mode. The zero role-switch count is expected because the implemented validation case uses fixed leader and follower identities for repeatability. The minimum separation remained positive, and both robot path lengths were recorded, confirming that the metrics logger observed the team-level behavior required by REQ-004.

The result should be interpreted within its intended scope. The simulation validates the traceability chain and the two-robot coordination behavior; it does not yet prove robustness across many environments, nor does it replace physical safety validation. The physical check further shows why this boundary is necessary: odometry and command interfaces were available, but lidar scan messages were not observed during the logged validation window. For this reason, the physical launch mode keeps direct motion disabled unless the operator explicitly enables it.

V. CONCLUSION AND DISCUSSION

This paper presented a SysML-to-ROS 2 workflow for a two-TurtleBot4 target-acquisition mission. The workflow connects requirement, behavioral, structural, software, hardware, and data-flow models to executable ROS 2 nodes, topics, launch files, DDS settings, and metrics. The implemented

TABLE 2. Detailed stable two-TurtleBot4 simulation validation results

Target case	Target (m)	Trials	Success	Completion time (s)	Path length (m)	Safety / spacing (m)
sim_t01	(1.422, 0.717)	3	3/3	17.337 ± 19.203	R1 0.473; R2 0.470; team 0.944	min 0.321; mean 0.524
sim_t02	(1.722, 0.717)	3	3/3	7.172 ± 1.607	R1 0.586; R2 0.195; team 0.781	min 0.347; mean 0.515
sim_t03	(1.572, 0.917)	3	3/3	5.003 ± 0.002	R1 0.353; R2 0.000; team 0.353	min 0.669; mean 0.669
sim_t04	(1.522, 1.117)	3	3/3	4.503 ± 0.503	R1 0.191; R2 0.000; team 0.191	min 0.659; mean 0.669
sim_t05	(1.722, 1.317)	3	3/3	4.002 ± 0.006	R1 0.278; R2 0.000; team 0.278	min 0.521; mean 0.522
All targets	5 cases	15	15/15 (100%)	mean 7.603 ± 8.928; median 5.004; range 3.997–39.509	R1 0.376 ± 0.146; R2 0.133 ± 0.231; team 0.509 ± 0.337; R2/R1 0.261 ± 0.463	global min 0.321; mean 0.580 ± 0.115; no observed contact

TABLE 3. Physical TurtleBot4 navigation-trial result template

Target case	Target (m)	Trials	Success	Completion time (s)	Path length (m)	Safety / spacing (m)
real_t01	(1.422, 0.717)	TBD	TBD	TBD	R1 TBD; R2 TBD; team TBD	min TBD; mean TBD
real_t02	(1.722, 0.717)	TBD	TBD	TBD	R1 TBD; R2 TBD; team TBD	min TBD; mean TBD
real_t03	(1.572, 0.917)	TBD	TBD	TBD	R1 TBD; R2 TBD; team TBD	min TBD; mean TBD
real_t04	(1.522, 1.117)	TBD	TBD	TBD	R1 TBD; R2 TBD; team TBD	min TBD; mean TBD
real_t05	(1.722, 1.317)	TBD	TBD	TBD	R1 TBD; R2 TBD; team TBD	min TBD; mean TBD
All targets	TBD	TBD	TBD	mean TBD; median TBD; range TBD	R1 TBD; R2 TBD; team TBD; R2/R1 TBD	global min TBD; mean TBD; contact events TBD

case study assigns `/robot1` as leader and `/robot2` as follower, publishes observable mission state, runs in simulation, and preserves the canonical `/robot1` and `/robot2` interfaces that can be connected to physical TurtleBot4 Lite robots through namespace or bridge adaptation.

The main quantitative validation remains limited to a two-robot target-acquisition task in the stable simulation mode. Within that boundary, 15 repeated trials over five target locations completed successfully and produced traceable CSV metrics. The physical deployment check confirmed the canonical odometry and command interfaces but did not observe lidar scan messages within the validation window, so physical navigation performance is not claimed. Future work will stabilize the physical sensing stream, connect the real TurtleBot4 robots through a safety-aware navigation stack on a larger validated map, and evaluate the same traceability process across more varied environments.

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